

# LOW-POWER 5V SUPPLY From Cells

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Obtaining 5V from AA or AAA cells is a difficult task in terms of high efficiency and a low component count. Classic boost converter circuits are challenging to implement in portable devices like wireless mice, remotes, etc. Presented here is a circuit that generates 5V from 1.5V or any single cell by using only three external components. The author's prototype assembled on a double-sided PCB is shown in Fig. 1.

Components required to build

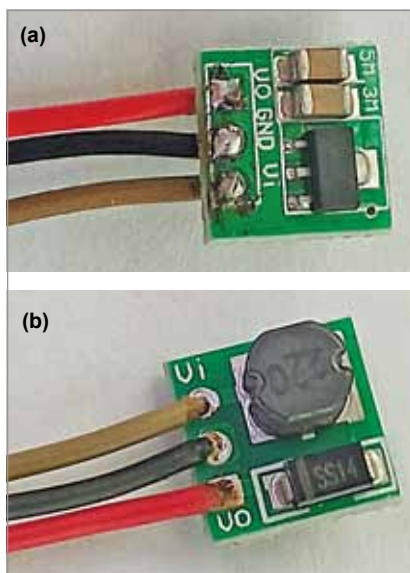


Fig. 1: Author's prototype: (a) solder side, (b) component side

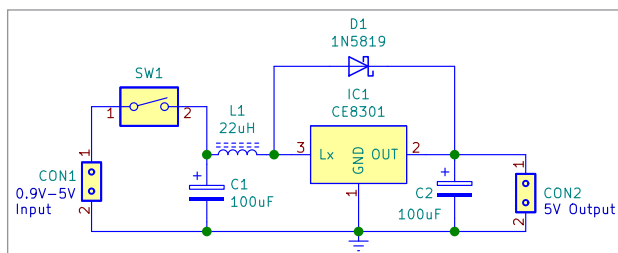


Fig. 2: Circuit diagram

this circuit are listed in bill of materials table.

## Circuit and working

Pulse frequency modulation (PFM) is a switching method normally used in many DC-DC voltage converters to improve efficiency at light loads. The circuit is built around PFM DC-DC boost converter chip CE8301 (IC1), an inductor 22 $\mu$ H (L1), a Schottky barrier diode 1N5819 (D1), and a few other components.

A single 1.5V cell is connected to connector CON1. Switch SW1 is used to turn on/off the circuit. When the switch is closed, IC1 and inductor L1 perform the step-up operation, and the 5V output voltage is available at CON2. Capacitor C2 is used to reduce the voltage ripple in the output voltage, while capacitor C1 acts as a buffer.

The CE8301 is a PFM step-up DC/DC converter IC with low supply current by CMOS process. Only three external components (inductor, Schottky diode, and output capacitor) are required to construct the boost converter. The converter's output voltage can be adjusted from 1.8V to 6.5V, 0.1V stepwise. The maximum operating frequency of the IC is typically 100kHz. It provides a simple-to-use, low-cost,

and space-saving solution. The IC has a low startup voltage of 0.9V and a guaranteed minimum current of 1mA. The IC CE8301 with a fixed 5V output is used in this circuit.

## BILL OF MATERIALS

| Components                | Quantity |
|---------------------------|----------|
| CE8301 (IC1)              | 1        |
| 1N5819 (D1)               | 1        |
| On-off switch (SW1)       | 1        |
| 22 $\mu$ H inductor (L1)  | 1        |
| 100 $\mu$ F, 16V (C1, C2) | 2        |
| 2-pin connector           | 2        |

## Construction and testing

Only a few components are used in this low-power supply circuit, so we have not designed its PCB. Its module is easily available in the market, so there is no need for design. However, an actual size single-sided or double-sided PCB can be used to assemble the circuit on the PCB.

Since the CE8301 IC is in an SMD package, solder it on the solder side of the PCB. Connect 1.5V across the connector CON1. Place the external components as close as possible to IC1 for optimal performance.

**EFY notes.** The ICs LTC3525-5, ME2108, and BL8530 can be used in place of IC1. The tantalum type capacitor is recommended for use as capacitor C2. The value of capacitor C2 should be above 10 $\mu$ F, and the voltage of the capacitor should be three times the output voltage of IC1. It is recommended to use a low ESR capacitor and a low-resistance inductor with a large current rating. **EFY**



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